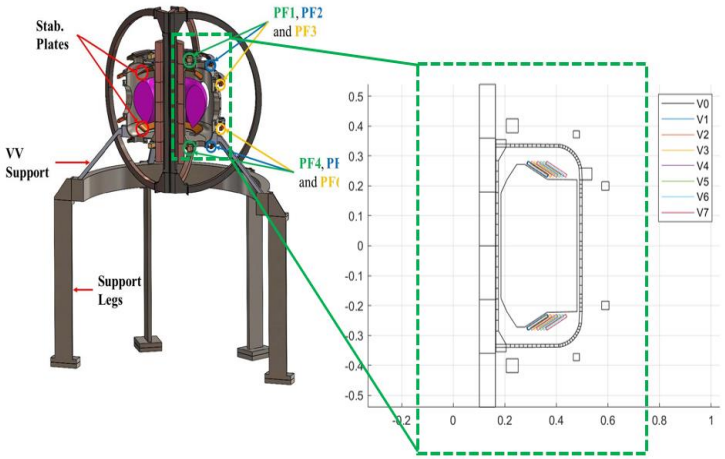
Analysis and Optimization of the position of Stabilizing Plates in the TRUST Experiment

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Co-Supervisor: PhD. Matteo Notazio/PhD. Mattia Scarpari

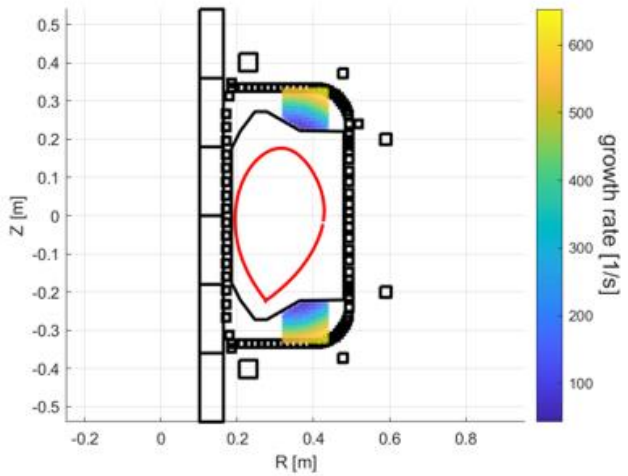
Objective: Analyse the position of the stabilizing plates to identify the best position to locate the plates within the Tuscia Research University Small Tokamak.



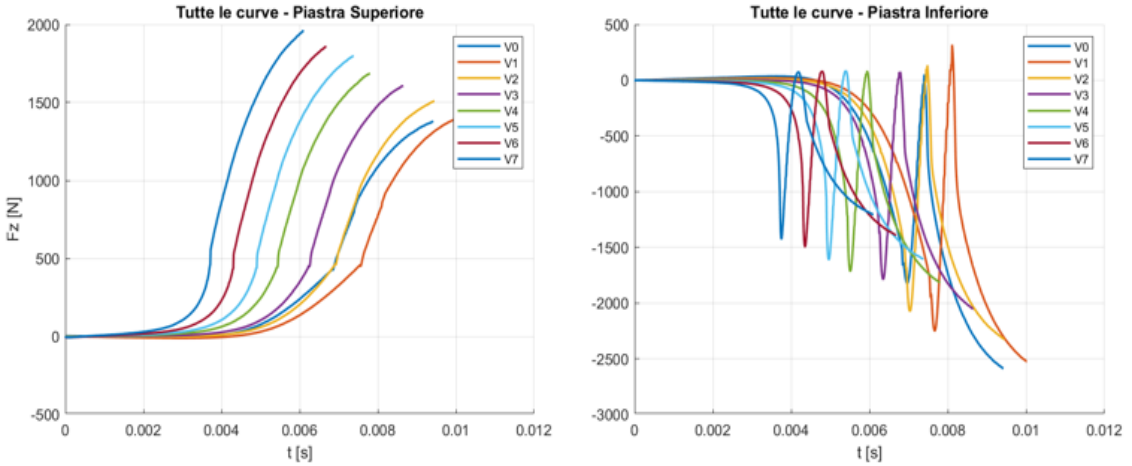
WORKFLOW

- GEO 2D
- AREA PER STABILIZZARE LE PIASTRE
- ANALISI 0D
- ANALISI 2D
- MODELLO RZIP/FIL.
- FORZA DI LORENTZ
- POSIZIONE OTTIMALE DELLE PIASTRE

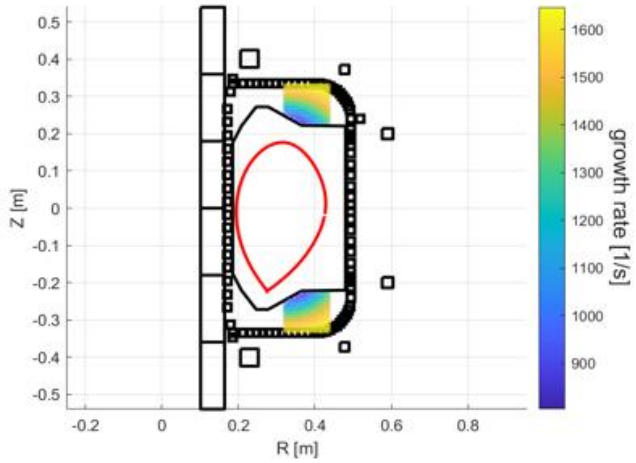
MATLAB: Modello Filamentare



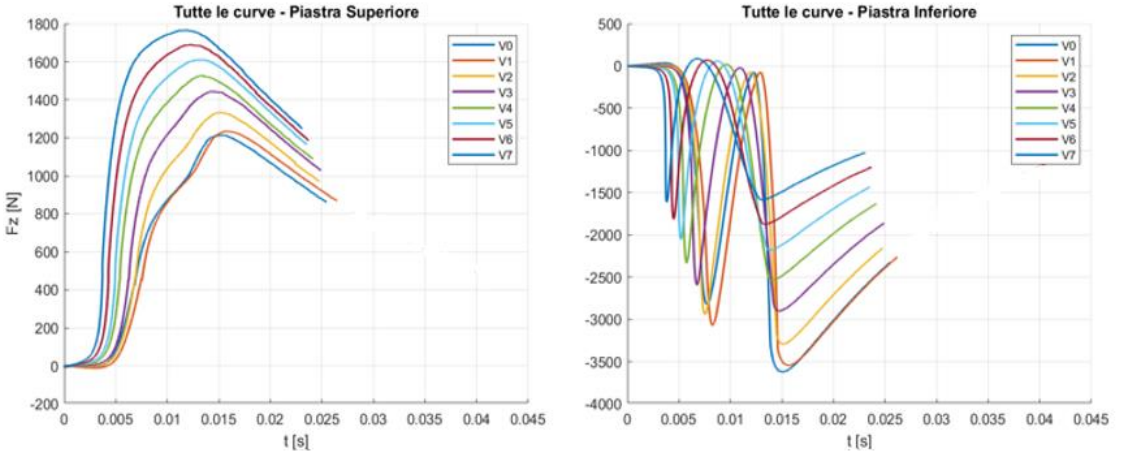
MAXFEA: Fast DVDE @ EOF



MATLAB: Modello RZIP



MAXFEA: Slow DVDE @ EOF



The results confirmed that stabilizing plate position significantly impacts the growth rate ("GR"). The RZIp model showed +23% GR increase when moving away from the first-wall compared to the selected V_0 position (respectively 987 s^{-1} with RZIp model and 118 s^{-1} with Filamentary model). Overall, V_0 remains the best compromise between plasma stability and plates loads, with forces of 2587 N (FAST) and 3622 N (SLOW) for the bottom plate and 1379 N (FAST) and 1214 N (SLOW) for the upper plate